NASA SPACE APPS CHALLENGE 2026 · FIELD SHIFT: ADAPTING FARMS WITH NASA DATA

Red Oasis

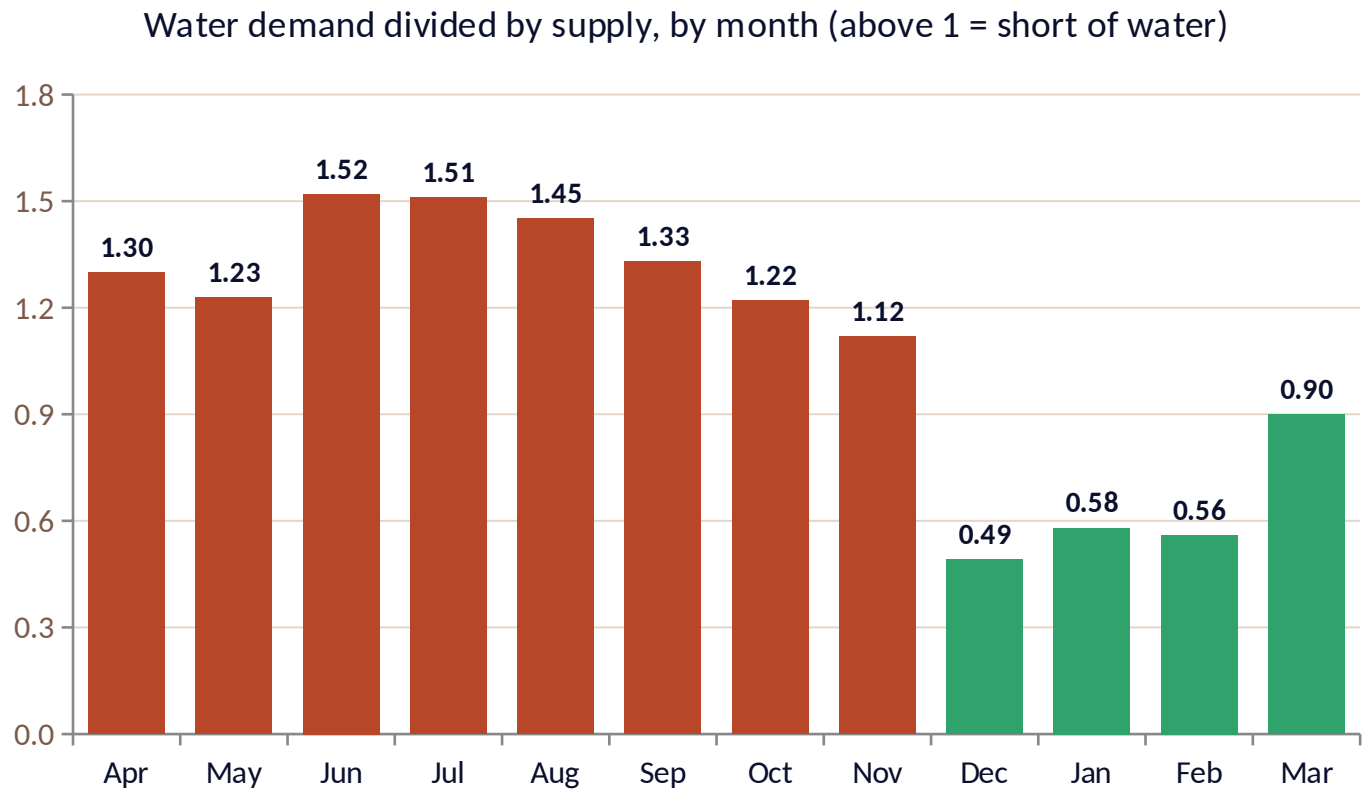
An irrigation decision-support prototype for date-palm oases

FROM THE FALAJ TO MARS

Team Red Oasis, Oman

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The Problem



1.12

demand/supply for the year: 3,620 mm needed, 3,235 mm supplied. Demand exceeds supply April to November; December to February has a surplus.

In space

On long Moon and Mars missions, food cannot rely on Earth, so every litre of water counts.

Source: Al-Ghafri (2006), observations from 2002-2003 (field research by Shareefa). Historical site-study values, not current measurements.

Our Solution

Red Oasis forecasts soil moisture, suggests when to irrigate and explains why. It uses NASA Earth observation data.

When to irrigate

One of three levels: irrigate today, monitor closely, or no irrigation needed. The rule is provisional.

WORKING

Why

Every forecast shows the factors behind it, using SHAP, in English or Arabic.

WORKING (EXPERIMENTAL)

Earth and Mars

The same decision logic is shown for a closed, water-recycling greenhouse.

CONCEPT

WORKING (EXPERIMENTAL) Trained model and SHAP explanation: working (experimental).

Pilot Site: Ad Dariz Area

Ad Dariz area

Ibri, Ad Dhahirah, Oman

CROP

Date palm (Phoenix dactylifera)

COORDINATES

23.30864 N, 56.61308 E

SITE

Ad Dariz agricultural area, Ibri



Image: Esri World Imagery (Vantor / WorldView-3, 8 May 2024). The project point is provisional.

Why Dariz?

- An arid oasis irrigated by a falaj, where date palm is the main crop.
- Published, site-specific evidence that irrigation demand exceeds supply.
- A real place for the advisor, instead of an invented farm.

Honest limits

The grove boundary still needs GIS or field confirmation. The Dariz study values are historical (2002-2003).

NASA Data

NASA source	Variable	Use in the model	Status
SMAP	Soil moisture	Forecast target and input	**Used (Dariz, 2024-2026)**
MODIS / VIIRS	Vegetation index (NDVI)	Vegetation condition (16-day composite)	**Used (Dariz, 2024-2026)**
GPM IMERG	Precipitation	Rain in the last 7 days	**Used (Dariz, 2024-2026)**
NASA POWER	Temperature and solar radiation	Air temperature and ET0	**Used (Dariz, 2024-2026)**

Regional, not field-level

A SMAP pixel is about 9 km wide, far larger than the Ad Dariz grove, so it gives regional context, not field-level truth.

Data period

1,004 daily rows from January 2024 to September 2026. A longer 2015 to 2026 record was also tested and did not change the conclusion.

How It Works

Pipeline

NASA data
SMAP, MODIS/VIIRS, GPM, NASA POWER

USED

→

Preprocessing
Daily table for the Dariz site

WORKING

→

Prediction model
Random Forest: soil moisture in 3 days

WORKING (EXPERIMENTAL)

→

SHAP explanation
Shows why the model gave its forecast

WORKING (EXPERIMENTAL)

→

Advisor app
English and Arabic interface

WORKING

Not done yet

Rule validation
The irrigation levels are a provisional rule, not validated advice.

PLANNED

Water-saving estimate
Planned by simulation, labelled as an estimate. No result yet.

PLANNED

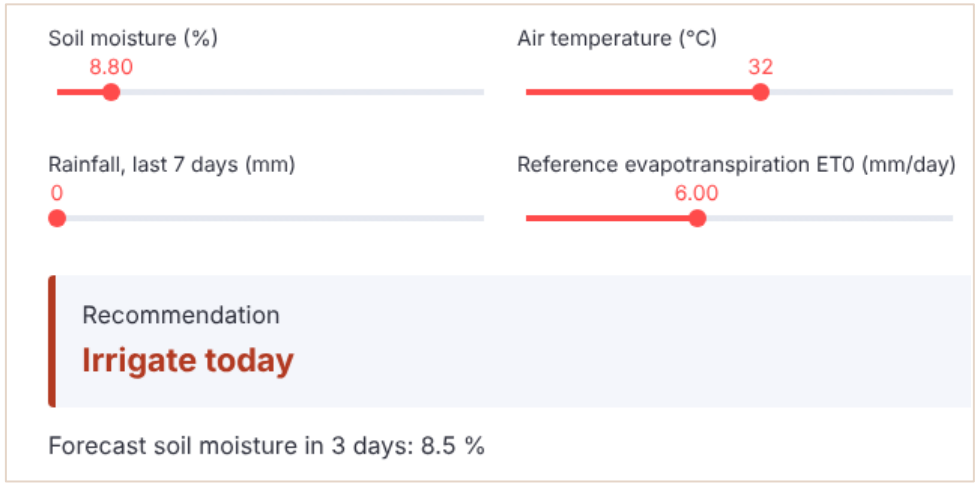
Field validation
No field measurements or grower logs yet.

PLANNED

Solid boxes are in the repository today; experimental ones do not reliably beat a no-change forecast. Dashed boxes are not done.

The Prototype

Dry case: soil moisture 8.8%, no rain
Result: Irrigate today (forecast in 3 days: 8.5%)



After-rain case: soil moisture 12.3%, 40 mm rain in 7 days
Result: No irrigation needed (forecast: 11.7%)

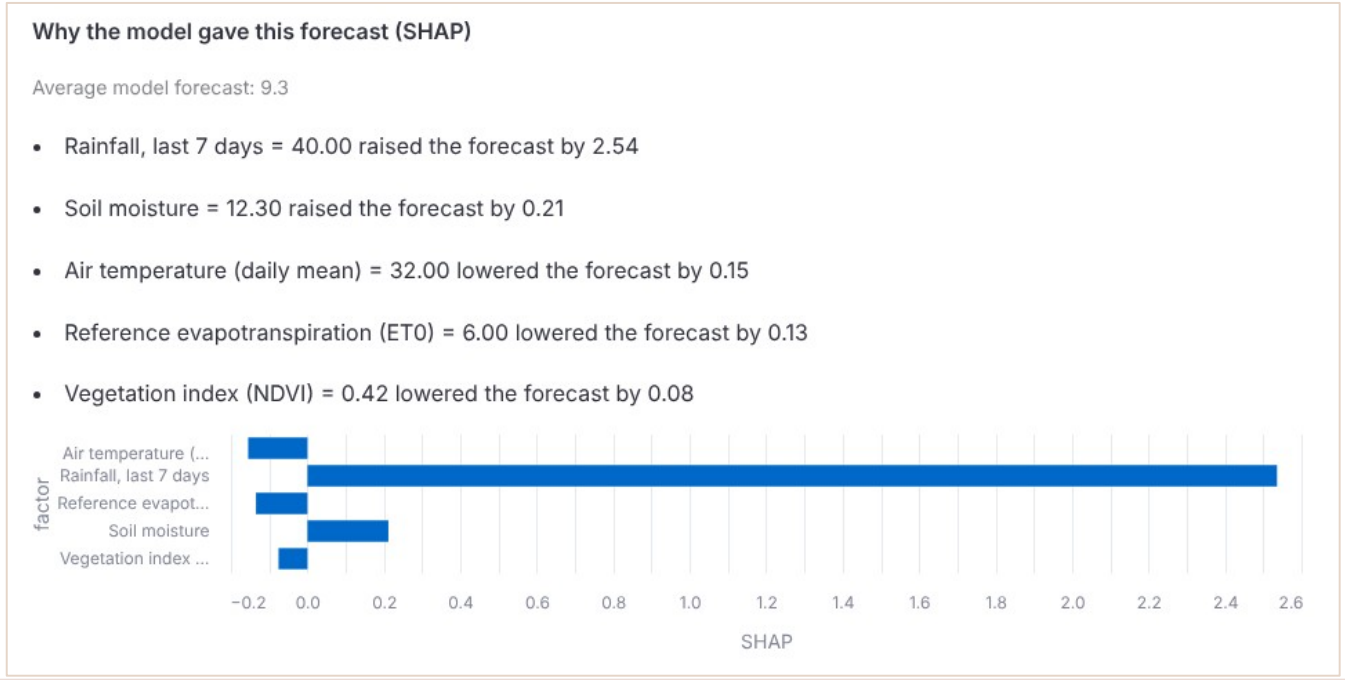


WORKING (EXPERIMENTAL)

Forecast error (mean absolute error, points of soil moisture). Final test, 122 days: model 1.20, no-change forecast 1.31. Cross-validation: model 1.18, no-change 1.03. The levels come from a provisional rule, not validated advice.

Explainable Advice

Why the model gave this forecast (SHAP), after-rain case



Bars to the right raised the forecast; bars to the left lowered it. SHAP shows how the model uses each input and is not proof of cause.

1.20

Model error, final test (122 days)

1.31

"No change" forecast, same test

1.18 vs 1.03

Cross-validation: model vs "no change"

WORKING (EXPERIMENTAL)

Better than "no change" on the final test only, so we call it experimental.

Mars Mode (Concept)



A concept, not a product

Water what-if

Tested at 50%, 75% and 100% make-up water: demand met 50%, 75% and 100%, shortfall 50%, 25% and 0%. The percentages are team assumptions, not NASA values.

Light and temperature

NASA ISS lettuce and plant-growth references, not validated for Mars or date palm.

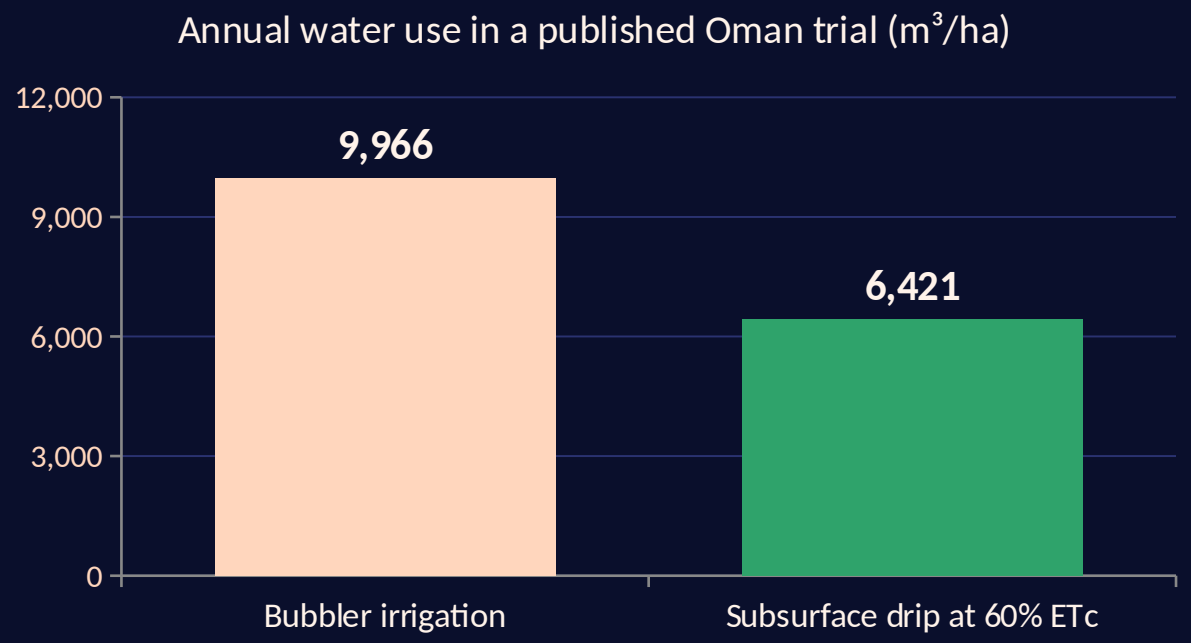
Evidence

NASA MarsOASIS is a lettuce greenhouse concept. It does not show that date palms can grow on Mars.

CONCEPT

Illustrative arithmetic only. It does not simulate a greenhouse.

Evidence and Next Steps



Published evidence, not our result

Yield was 79.0 kg per palm with bubblers and 78.3 kg with drip, with no significant difference reported. In the app, the farmer sets the yield loss they would accept.

Source: Dhehibi et al. (2022); Al Kamil and Al Wafi, 2015-2016. A different site from Dariz and not a Red Oasis result.

Next steps

PLANNED

1. Test the rule with growers and field data
2. Calibrate palm thresholds in the field
3. Estimate water saving by simulation, labelled as an estimate
4. Confirm the grove boundary

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